

Exercise 1: Control Structures

Scenario 1: Apply 1% Discount to Loan Interest Rate for Customers Above 60

```
DECLARE
  CURSOR customer_cursor IS
    SELECT c.CustomerID, c.DOB, l.LoanID, l.InterestRate
    FROM Customers c
    JOIN Loans l ON c.CustomerID = l.CustomerID;

  age NUMBER;
BEGIN
  FOR rec IN customer_cursor LOOP

    age := FLOOR(MONTHS_BETWEEN(SYSDATE, rec.DOB) / 12);

    IF age > 60 THEN
      UPDATE Loans
      SET InterestRate = InterestRate - 1
      WHERE LoanID = rec.LoanID;

      DBMS_OUTPUT.PUT_LINE('Interest updated for Customer ID: ' || rec.CustomerID);
    END IF;

  END LOOP;

  COMMIT;
END;
/
```

Scenario 2: Promote Customers to VIP Status

```
ALTER TABLE Customers
ADD IsVIP VARCHAR2(5);
BEGIN

  FOR rec IN (SELECT CustomerID, Balance FROM Customers) LOOP

    IF rec.Balance > 10000 THEN

      UPDATE Customers
      SET IsVIP = 'TRUE'
      WHERE CustomerID = rec.CustomerID;
```

```
        END IF;

    END LOOP;

    COMMIT;

END;
/
\
```

Scenario 3: Loan Reminder (Due Within 30 Days)

```
BEGIN

    FOR rec IN (

        SELECT c.Name,
               I.LoanID,
               I.EndDate
        FROM Customers c
        JOIN Loans I
        ON c.CustomerID = I.CustomerID
        WHERE I.EndDate BETWEEN SYSDATE AND SYSDATE + 30

    ) LOOP

        DBMS_OUTPUT.PUT_LINE(
            'Reminder: Loan '
            || rec.LoanID
            || ' for '
            || rec.Name
            || ' is due on '
            || TO_CHAR(rec.EndDate,'DD-MON-YYYY')
        );

    END LOOP;

END;
/
```

Exercise 3: Stored Procedures

Scenario 1: Process Monthly Interest

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
IS
BEGIN

    UPDATE Accounts
    SET Balance = Balance + (Balance * 0.01)
    WHERE AccountType = 'Savings';

    COMMIT;

END;
/
BEGIN
    ProcessMonthlyInterest;
END;
/
```

Scenario 2: Update Employee Bonus

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(

    p_department IN VARCHAR2,
    p_bonus IN NUMBER

)

IS
BEGIN

    UPDATE Employees
    SET Salary = Salary + (Salary * p_bonus / 100)
    WHERE Department = p_department;

    COMMIT;

END;
/
BEGIN
```

```
UpdateEmployeeBonus('IT',10);  
END;  
/
```

Scenario 3: Transfer Funds

```
CREATE OR REPLACE PROCEDURE TransferFunds(
```

```
    p_fromAccount IN NUMBER,  
    p_toAccount IN NUMBER,  
    p_amount IN NUMBER
```

```
)
```

```
IS
```

```
    v_balance NUMBER;
```

```
BEGIN
```

```
    SELECT Balance  
    INTO v_balance  
    FROM Accounts  
    WHERE AccountID = p_fromAccount;
```

```
    IF v_balance >= p_amount THEN
```

```
        UPDATE Accounts  
        SET Balance = Balance - p_amount  
        WHERE AccountID = p_fromAccount;
```

```
        UPDATE Accounts  
        SET Balance = Balance + p_amount  
        WHERE AccountID = p_toAccount;
```

```
        COMMIT;
```

```
        DBMS_OUTPUT.PUT_LINE('Transfer Successful');
```

```
    ELSE
```

```
        DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
```

```
END IF;
```

```
END;
```

```
/
```

```
BEGIN
```

```
    TransferFunds(1,2,500);
```

```
END;
```

```
/
```